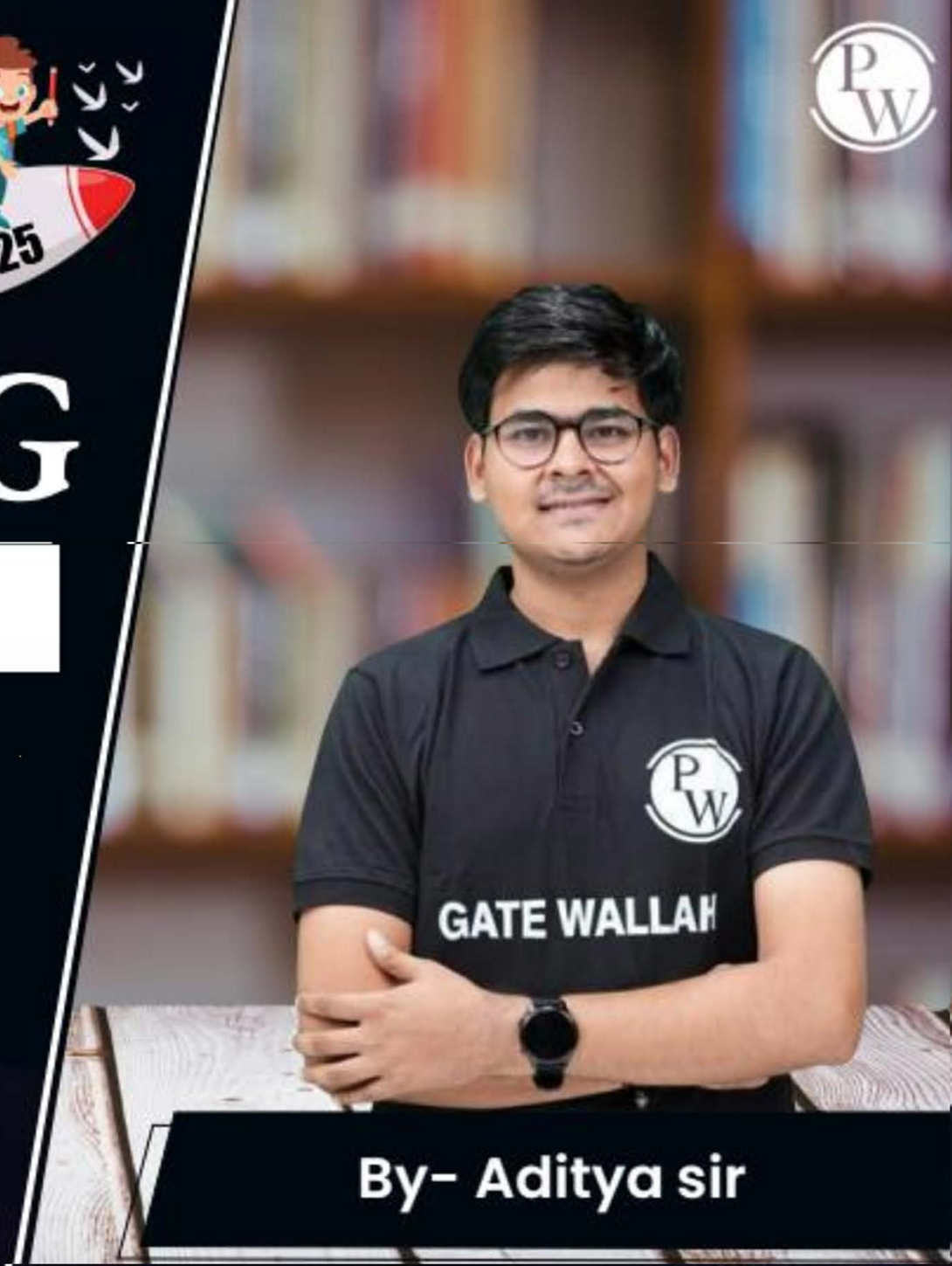
CS & IT ENGINEERING



Algorithms

Greedy Method

Lecture No.- 01



By- Aditya sir

Recap of Previous Lecture



Topic Master's method

Topic Practice + PYQs

Topics to be Covered



- **Topic** Practice + PYQs
- **Topic** Greedy Algorithm

ABOUT ME

Hello, I'm Aditya.

1. Represented college as the first Google DSC Ambassador.
2. The only student from the batch to secure an internship at Amazon.
3. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis8. Completed my Masters with an overall GPA of 9.36/108. Joined Dream11 as a Data Scientist.9. Have mentored around 10k working professions in field of Data Science and Analytics10. Have been mentoring GATE aspirants to secure a great rank in limited time.11. Have got around 27.5K followers on Linkedin where I share my insights and guide students and professionals.



By- Aditya sir

[MCQ]

948

721.

#Q. For Constant $a \geq 1$ and $b > 1$, consider the following recurrence defined on the non-negative integers:

$$T(n) = a.\,T\left(\frac{n}{b}\right) + f(n)$$

Ans :- B

Which one of the following options is correct about the recurrence T(n)?

A. If $f(n)$ is $\theta\left(n^{\log_b(a)}\right)$ then $T(n)$ is $\theta\left(n^{\log_b(a)}\right)$

B. If $f(n)$ is $O\left(n^{\log_b(a)-e}\right)$ for some $e > 0$, then $T(n)$ is $\theta\left(n^{\log_b(a)}\right)$ → Case 1

C. If $f(n)$ is $\frac{n}{\log_2(n)}$ then $T(n)$ is $\theta(n \log_2(n))$

D. If $f(n)$ is $n \log_2(n)$, then $T(n)$ is $\theta(n \log_2(n))$

[MCQ]

PYQ

#Q. For parameters a and b, both of which are $\omega(1)$, ~~$T(n) = T(n^{1/a}) + 1$~~, and T(b) = 1. Then T(n) is :

$T(n) = T(n^{1/a}) + 1$

↳ Solve using Bach-Substi:

58%

A $O(\log_2 \log_2 n)$

B $O(\log_b \log_a n)$

C $O(\log_a \log_b n)$

D $O(\log_{ab} n)$

$T(n) = T(n^{1/a}) + 1$

$T(b) = 1$

$T(n) = T(n^{1/a}) + 1$ — (1)

$T(n^{1/a}) = T(n^{1/a \times \frac{1}{a}}) + 1$

$T(n^{1/a}) = T(n^{1/a^2}) + 1$

$T(n) = T(n^{1/a^2}) + 1 + 1$

$T(n) = T(n^{1/a^2}) + 2$ — (2)

$T(n^{1/a^2}) = T(n^{1/a^3}) + 1$

$T(n) = T(n^{1/a^3}) + 3$ — (3)

In general,

$$T(n) = T(n^{1/a^k}) + k \quad \text{——} (4)$$

B.C $\Rightarrow$ $T(b) = 1$

$$n^{1/a^k} = b$$

$$\frac{1}{a^k} \log_2 n = \log_2 b \Rightarrow$$

$$a^k = \frac{\log_2 n}{\log_2 b} = \log_b n$$

$$\boxed{k = \log_a (\log_b n)}$$

$$T(n) = T(n^{1/a^k}) + k$$

$$T(n) = T(b) + k$$

$$T(n) = 1 + \log_a(\log_b n)$$

$$T(n) = O(\log_a(\log_b n))$$



Topic : Greedy Algorithm

49%

Always

(V. Imp)

QS: Middle element as pivot

Worst case TC = ?

A) $O(n)$

B) $O(n^2)$

C) $O(n \log n)$

D) $O(\sqrt{n})$

mis-concept



$TC = O(n \log n)$



Topic : Greedy Algorithm

Before partition:



After partition:



(V. Imp)

- "Divide happens as per where the pivot get placed after partition."

  Hence, worst case : $O(n^2)$



Topic : Greedy Algorithm

In QS, if median is always selected as pivot then the worst case TC is?

Median → middle element in sorted order

A) O(n)

B) O(n²)

C) ~~O(nlogn)~~

D) O(n!)

TC = O(n²)

↳ wrong



Topic : Greedy Algorithm

Example:

10, 2, 9, 15, 13

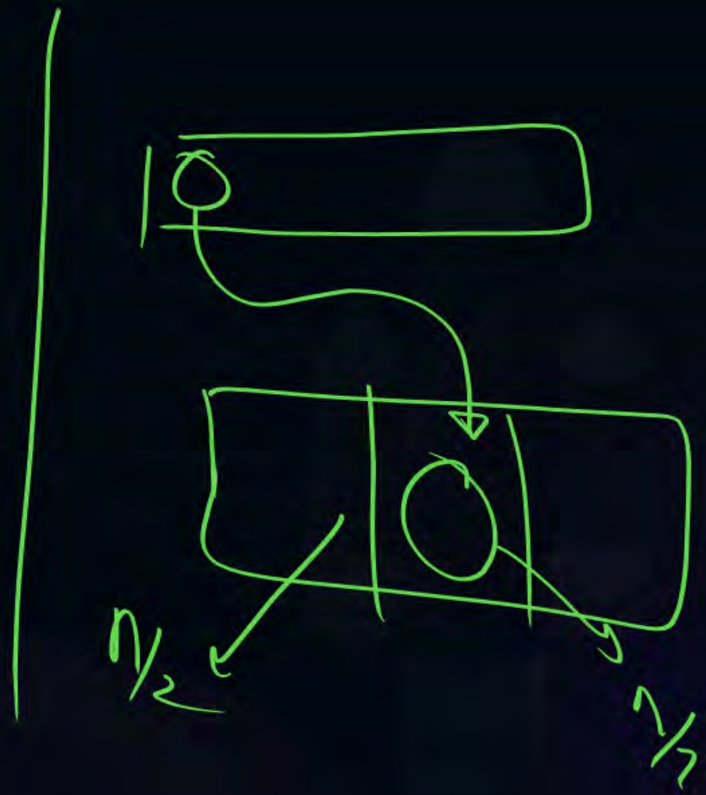
Median:

2, 9, 10, 13, 15

sort

pivot → 10

median → 10





Topic : Greedy Algorithm

Time Complexity:

$$T(n) = T(n/2) + T(n/2) + O(n) + O(n)$$

$$= O(n \log n)$$

median

partition

[MCQ]

#Q. When $n = 2^{2k}$ for some $k \geq 0$, then recurrence relation

$T(n) = \sqrt{2}T(n/2) + \sqrt{n}, \qquad T(1) = 1$

Evaluates to :

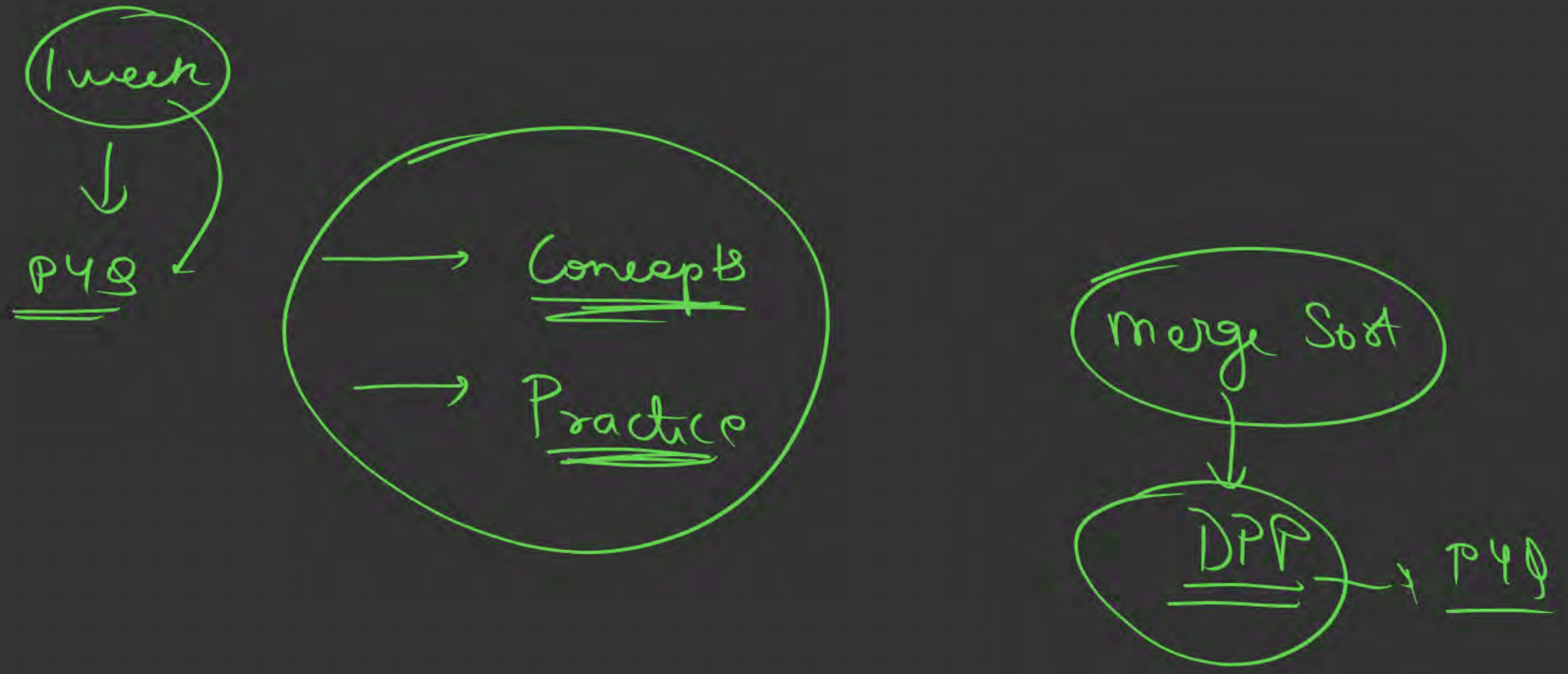
Ans: $\Theta(\sqrt{n}\log n)$

A $O(\log_2 \log_2 n)$

B $O(\log_b \log_a n)$

C $O(\log_a \log_b n)$

D $O(\log_{ab} n)$


1 week
P4S
Concepts
Practice
merge Sort
DPP
+ P4S

$$T(n) = \sqrt{2}\,T(n/2) + \sqrt{n}$$

$$T(n) = aT(n/b) + F(n)$$

$a = \sqrt{2}$
$b = 2$
$f(n) = \sqrt{n}$ } ✓

$$\log_b a = \log_2(\sqrt{2}) =$$

$$= (1/2)$$

Case 1 :- Is $\sqrt{n} = O(n^{1/2 - \varepsilon})$, $\varepsilon > 0$?

$\longrightarrow$ No

$\longrightarrow$ fails

Case 2 :- Is $\sqrt{n} = \Theta(n^{1/2} * (\log n)^k)$

1) $k \geq 0$ $\sqrt{n} = \Theta(\sqrt{n}(\log n)^0)$

$\sqrt{n} = \Theta(\sqrt{n})$, $k = 0$

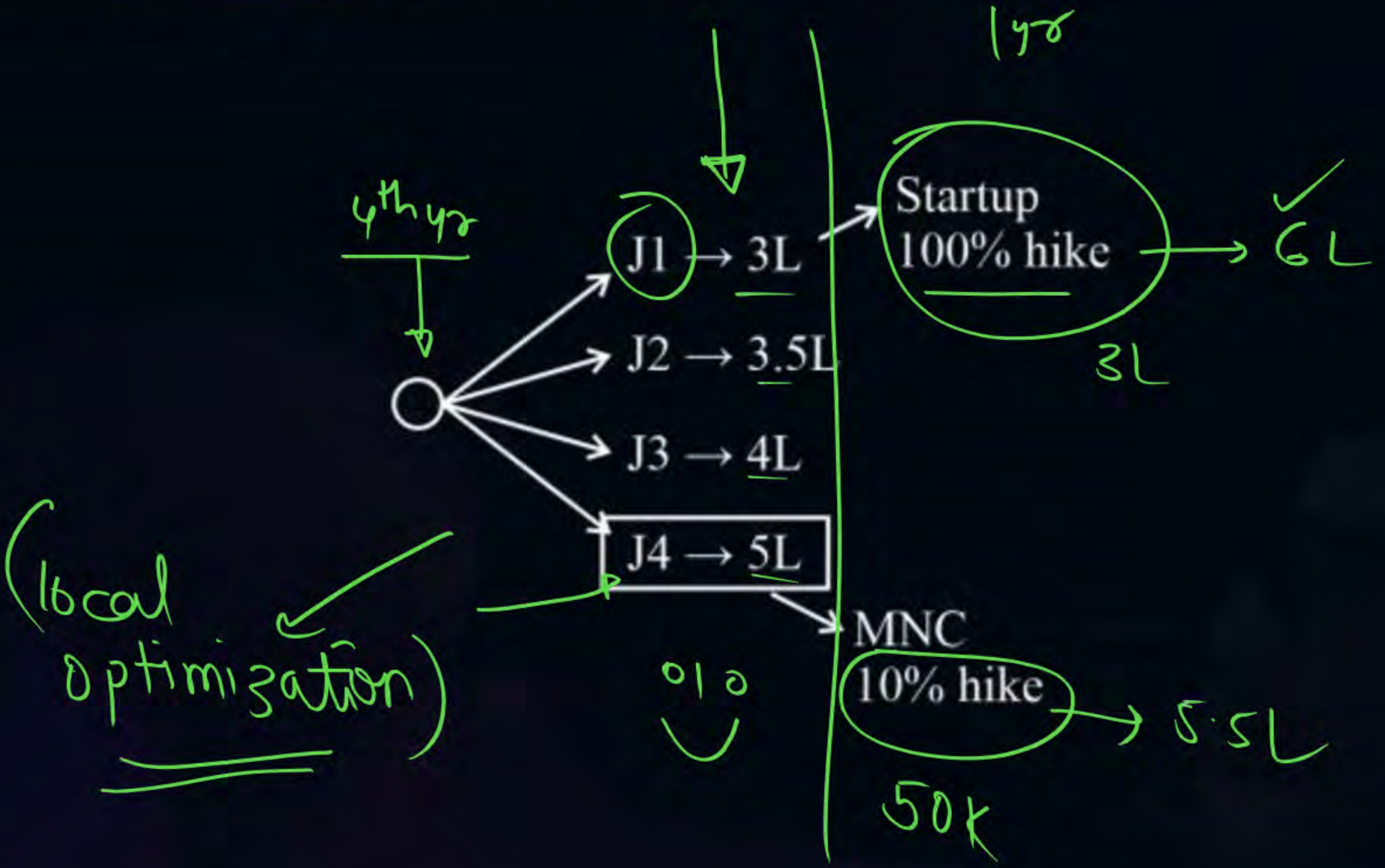
True

$T(n) = \Theta(n^{1/2} * \log n^{0+1}) = \Theta(\sqrt{n} \log n)$

{Greedy Algorithms}

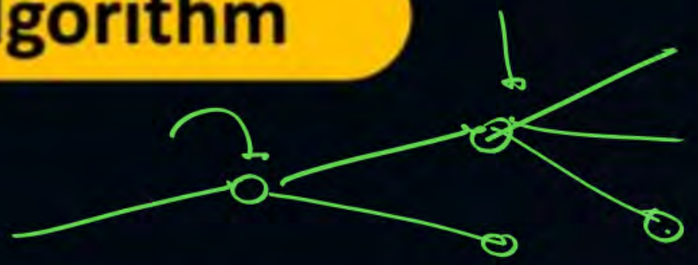
Topic : Greedy Algorithm

Idea:





Topic : Greedy Algorithm



Steps:

(1) Used for problems whose solutions are viewed as a set/sequence of decisions.

(2) These decisions are made in a step-wise manner.

(3) At each step, out of all the available options, gradually select these option, which satisfies the criteria of the problem.

(V. imp)



Topic : Knapsack Problem



Description:

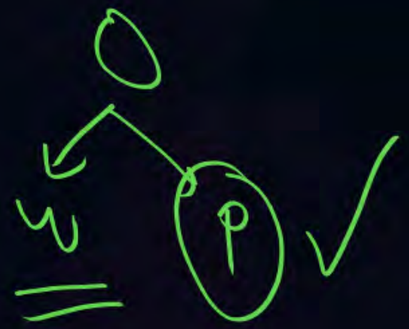
(1) Given a Knapsack bag of capacity 'm'.

(2) Given n objects:

Object (i) → Weight (W_i)

Object (i) → Profit (P_i)

Object (i) → Decision (X_i) →

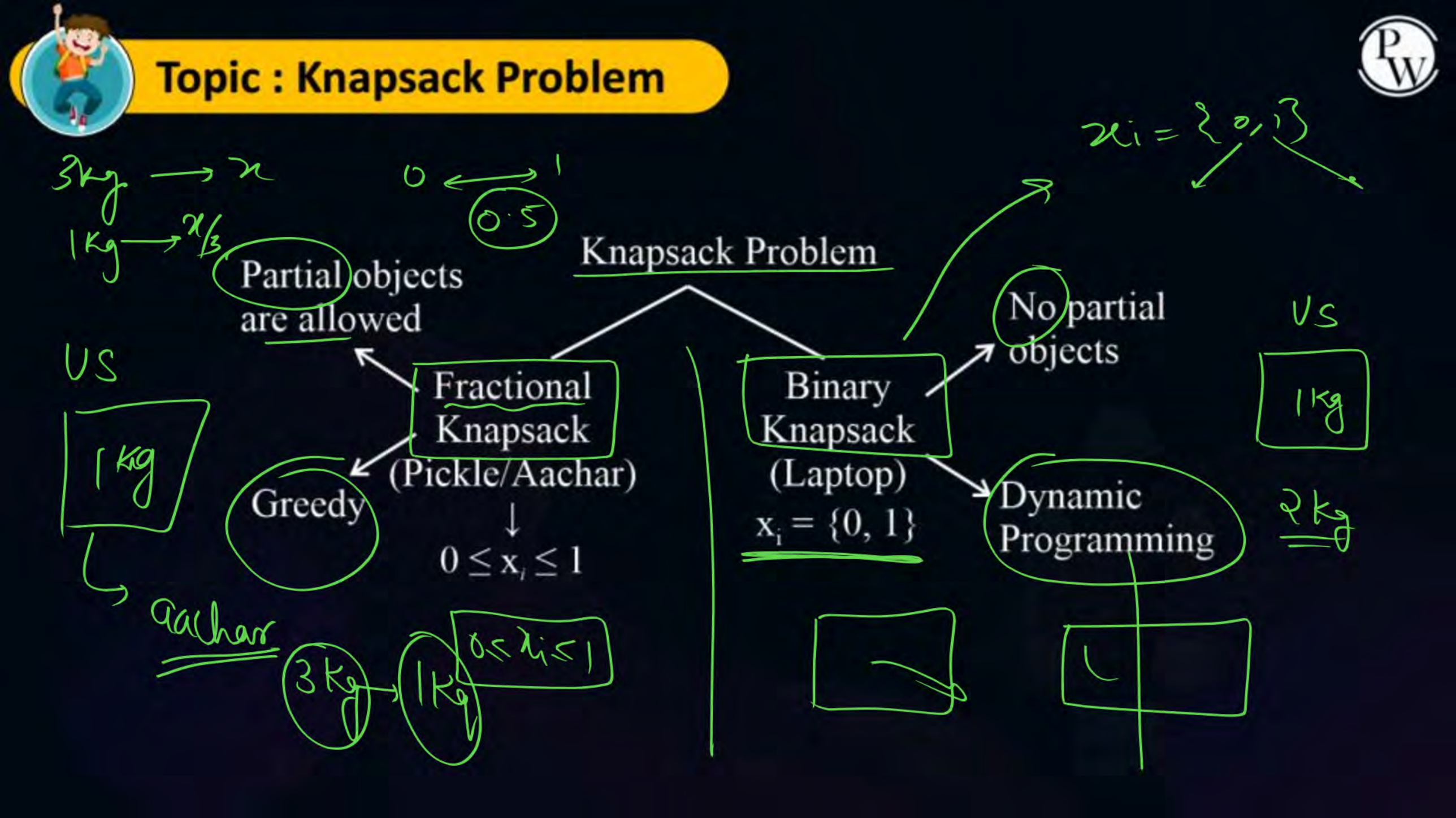




Topic : Knapsack Problem

- You have to maximize the profit subject to the criteria that the total weight put into the knapsack does not exceed its capacity (M).

Topic : Knapsack Problem



Topic : Knapsack Problem

eg

$O_1 \rightarrow P_1$	W_1	X_1
$O_2 \rightarrow P_2$	W_2	X_2
$O_3 \rightarrow P_3$	W_3	X_3

O_i — $P_i \times x_i$

$\Sigma P_i \rightarrow$ maximize

$\Sigma x_i * P_i \rightarrow$ Maximize

Subject to criteria:

$\Sigma x_i * W_i \leq M$

P_i
0

$x_i = 1/2$

1kg → 100

0.5kg → 50



Topic : Knapsack Problem



Decision x_i on object O_i

Profit: $x_i * P_i$

Weight: $x_i * W_i$



Topic : Knapsack Problem

Example:

Fractional Knapsack:

No. of objects, n = 3

Knapsack Capacity, M = 20

$[w_1 \ w_2 \ w_3] = [18, 15, 10]$

$[p_1 \ p_2 \ p_3] = [25, 24, 15]$

o/p

o/p int

2 8.2

Task 1 :- Greedy about Profit



Topic : Knapsack Problem

Aim: Maximize Profit:

Prefer objects with max Profit:

P_i :	25	24	15
W_i :	18	15	10
X_i :	1	$\frac{2}{15}$	0

$20 > 18$

$20 - 18 = 2$

$2 - 2 = 0$

$x_2 = \frac{2}{15}$

15 → 24

$1 \rightarrow \frac{24}{15}$

$2 \rightarrow \frac{24}{15} \times 2$

$25 + 24 \times \frac{2}{15} + 0 \times 15$

$25 + 3.2$

$= 28.2$



Topic : Knapsack Problem

$M = 20$

$18 < 20$

$M = 20 - 18 = 2$

$M = 2 - 2 = 0$

Hence Total profit by this approach

$$= \sum_{i=1}^{3} P_i * x_i$$

$$= P_1 \times x_1 + P_2 \times x_2 + P_3 \times x_3$$

$$= 28.2$$

Task 2 :- Greedy about weight

(min wt)

int



Topic : Knapsack Problem

Approach 2:

Greedy about weight:

P_i :	25	24	15
W_i :	18	15	10
X_i :	0	$\frac{10}{15}$	1

$25 \times 0 + 24 \times \frac{10}{15} + 15 \times 1$

$= 0 + 24 \times \frac{2}{3} + 15 = 31$

$20 > 10$

$20 - 10 = 10$

$10 < 15 \rightarrow$

$\frac{10}{15} \rightarrow 0$



Topic : Knapsack Problem

$M = 20$

$M = 20 - 10 = 10$

$M = 10 - 10 = 0$

Hence Total profit by this approach

$$= \sum_{i=1}^{3} P_i * x_i$$

$$= P_1 \times x_1 + P_2 \times x_2 + P_3 \times x_3$$

$$= 31$$

[Depends on data]

O_1

O_2

O_3

3rd Approach : Universal



Topic : Knapsack Problem

Approach 3:

Combines both logic (optimal):

	w_i	P_i
$01 \rightarrow 1$		P_1
$02 \rightarrow 1$		P_2
$03 \rightarrow 1$		P_3

↳ (proper max Profit)



Topic : Knapsack Problem

Weight adjusted Profits:

Profit per unit weight: P_i/W_i

$W_1 \rightarrow P_1$ $\quad$ $1 \rightarrow P_1/W_1$

$W_2 \rightarrow P_2$ $\quad$ $1 \rightarrow P_2/W_2$

$W_3 \rightarrow P_3$ $\quad$ $1 \rightarrow P_3/W_3$

$W \rightarrow P$

$1 \rightarrow P/W$



Topic : Knapsack Problem

Example:

Fractional Knapsack:

No. of objects, n = 3

Knapsack Capacity, M = 20

$[w_1\ w_2\ w_3] = [18, 15, 10]$

$[p_1\ p_2\ p_3] = [25, 24, 15]$

$20 - 15 = 5 < 10$

$5 - 5 = 0$

$O_1 \rightarrow \frac{25}{18} = 1.38$ — 3

$O_2 \rightarrow \frac{24}{15} = 1.6$ — 1 $x_2 = 1$

$O_3 \rightarrow \frac{15}{10} = 1.5$ — 2

$x_i \quad [0, 1, \frac{5}{10}]$

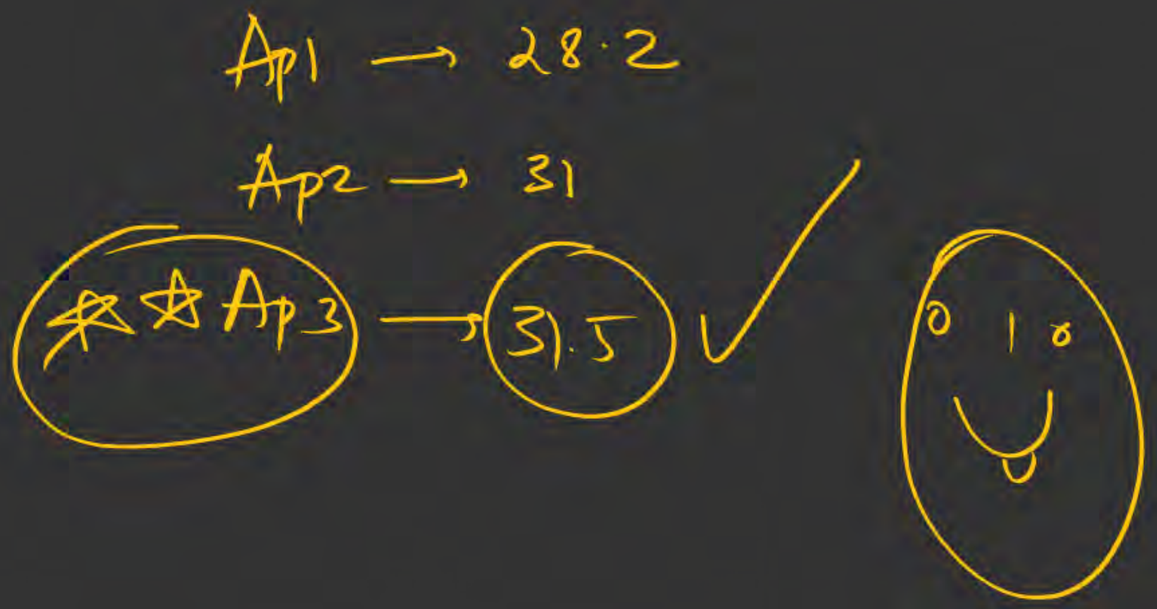
$$x_i = [0, 1, 0.5]$$

$$\text{Profit} = \sum R_i x_i$$

$$= 25 \times 0 + 24 \times 1 + 15 \times 0.5$$

$$= 24 + 7.5$$

$$= \boxed{31.5}$$ ✓


Ap1 → 28.2
Ap2 → 31
Ap3
31.5

[NAT]

#Q. Find max/optimal solution to the fractional knapsack problem:

Given: n = 7, M = 15

H.W

Object	Weight	Profit
01	4	18
02	1	3
03	3	5
04	7	7
05	1	6
06	5	15
07	2	10



THANK - YOU